

Sagar Shukla

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PROFILE

Computer Engineering graduate and M.Tech AI & ML student at COEP Technological University, with hands-on experience in Python, computer vision, deep learning, and Generative AI applications.

EDUCATION

COEP Technological University

Pune, India

Mtech in Artificial Intelligence and Machine Learning

2025 – present

- CGPI:- 8.68/10

Fr. Conceicao Rodrigues Institute of Technology, Mumbai University

Navi Mumbai, India

Bachelor of Engineering in Computer Engineering

2021 – 2025

- CGPI:- 7.97/10
- Specialization in Artificial Intelligence and Machine Learning.

SKILLS

Languages: Python , C , C++

Skills: Data Science, ETL, Data Visualization, Machine Learning, Computer Vision, Generative AI, Generative Adversarial Networks (GANs), Retrieval-Augmented Generation (RAG), Natural Language Processing

EXPERIENCE

Rajasthan Police

AI Intern

2/2024 – 7/2024

- Optimized AI-powered threat detection for 133 CCTV cameras, increasing incident response by 15%.
- Developed a deployable POC with over 5 use cases, including mobile IP cameras for threat detection.

Computer Society of India FCRIIT

Assistant Technical Coordinator

7/2023 – 5/2024

- Developed and maintained the CSI website, and organized technical events and coding competitions.
- Designed coding contest problems, managed evaluations, and ensured smooth execution of technical activities.

PUBLICATIONS

Chandresh Menon, Neha Narewadi, Sagar Shukla, Sebastian Veg, and Mrs. Shagufta, "Autonomous Criminal Identification System", GRENZE International Journal of Engineering and Technology, 2024 [pg2300-2305]

ACHIEVEMENTS

- Best Project Award, PCUBE 2025, Fr. Conceicao Rodrigues Institute of Technology.
- 2nd Rank in Hack Hurricane 1.0, a 48-hour national hackathon by PHICSIT.
- National Level Competitor, Srinivasa Ramanujan Mathematical Competition.
- Speaker at the AIML (Artificial Intelligence and Machine Learning) training session in college, presenting on basic algorithms of machine learning.

PROJECTS

SkySentinel: IoV-Integrated Autonomous Drone Swarm

6/2024 – 5/2025

- Developed an autonomous multi-drone surveillance system integrating **YOLOv11-Pose** for real-time suspicious activity recognition using human keypoints.
- Designed a **Dual Granularity GAT** with Transformer-based temporal modeling for accurate multi-person action classification.
- Enabled secure 4G-based MAVLink communication via VPN and **automated area division** using Voronoi Tessellation.
- Built a **PyQt6-based GCS** with Leaflet.js for real-time drone tracking and monitoring.
- Achieved high accuracy on NTU RGB+D 120 dataset.
- Won Best Project title at **PCUBE** Intra-Department Competition.

SME-PLUG — Hot-Swappable Subject Matter Expert Plugin

2/2026 – Present

- Built an **AI middleware plugin** that injects expert reasoning and verified citations into LLMs to reduce hallucinations.
- Designed a **3-layer domain router** (Regex → LLM → Embeddings) achieving **<500 ms classification latency**.
- Implemented a **hybrid RAG pipeline** using HyDE, BM25 + ChromaDB retrieval, RRF fusion, and cross-encoder reranking.
- Developed **“DNA Capsules”** — hot-swappable domain modules loadable without retraining the base model.
- Built a **FastAPI server** with **LangChain** and **CrewAI integrations**, with **hallucination detection** and **confidence scoring**.

Autonomous Criminal Identification System

6/2023 – 5/2024

- Designed a real-time criminal identification system using **YOLOv8**, **ResNet-50**, and **DeepSORT** for accurate human detection and tracking in public surveillance videos.
- Implemented **anomaly detection with 95%+ accuracy**, triggering alerts and storing incidents with timestamped video evidence.
- Built a **Django-based interface** for visualizing detections and enabling remote access for authorized personnel.
- **Published** the project *"Autonomous Criminal Identification System"* in the **GRENZE International Journal of Engineering and Technology**.

NetraGuard - Suspicious Activity Recognition

4/2024 – 6/2024

- **Multi-stage Suspicious Activity Recognition System** combining video classification and object detection.
- Object Detection using **YOLOv8** and tracking done using **DeepSORT**.
- Video classification using **ResNET-50**, achieved an accuracy of **97%**.
- EasyOCR for extracting text related information from the frames.
- Primarily trained and evaluated on the **UCF101 dataset**, enabling recognition of diverse human actions.

Government Scheme Explainer Chatbot

1/2026 – 2/2026

- Built a **government scheme explainer chatbot** using **LLaMA-3.1 8B Instruct** for social-good use cases.
- Achieved **80%+ response accuracy** with low-latency inference using **quantized LLM deployment**.
- Evaluated instruction-following, factual correctness, and real-world usability of LLM responses.